

Discrete Mathematics and Its Applications



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University of West England Phenikaa Campus

UCFF-30-1 Foundation of Computing - Graph

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Learning Objectives

1. Graph and Graph Models
2. Graph Terminology
3. Graph Representations

Introduction

- ❑ Discrete structures consisting of vertices and edges that connect these vertices
- ❑ Different kinds of graphs
- ❑ Problems in almost every conceivable discipline can be solved using graph models

10.1 Graphs and Graph Models

DEFINITION 1

A *graph* $G = (V, E)$ consists of V , a nonempty set of *vertices* (or *nodes*) and E , a set of *edges*. Each edge has either one or two vertices associated with it, called its *endpoints*. An edge is said to *connect* its endpoints.

Remark: The set of vertices V of a graph G may be infinite. A graph with an infinite vertex set or an infinite number of edges is called an **infinite graph**, and in comparison, a graph with a finite vertex set and a finite edge set is called a **finite graph**. In this book we will usually consider only finite graphs.

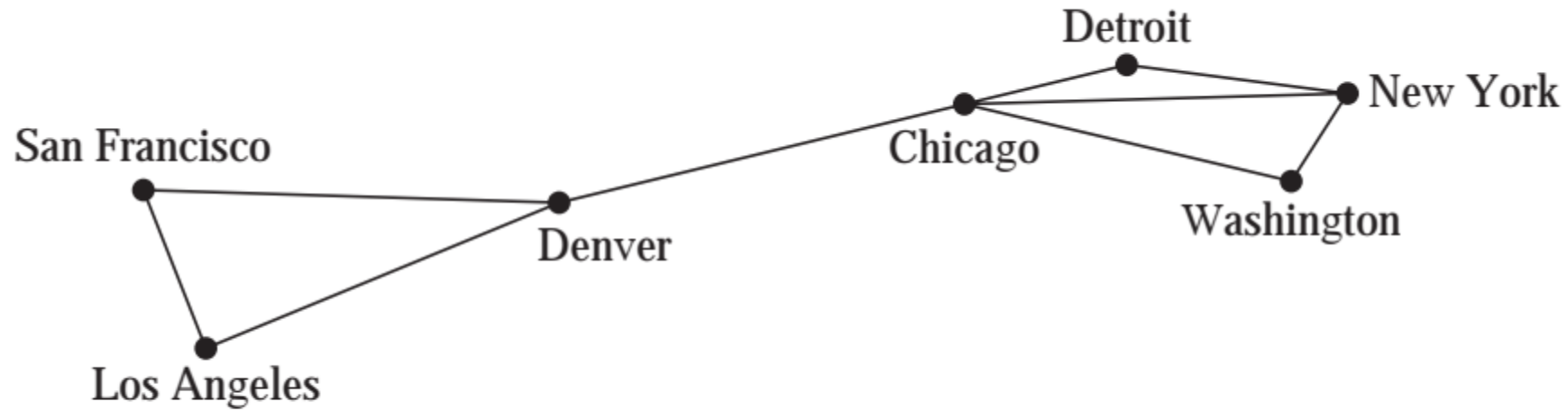


FIGURE 1 A Computer Network.

- ❑ A network is made up of data centers and communication links between computers.
- ❑ Represent the location of each data center by a point and each communications link by a line segment, as shown in Figure 1
- ❑ Can be modeled using a graph in which the vertices of the graph represent the data centers and the edges represent communication links

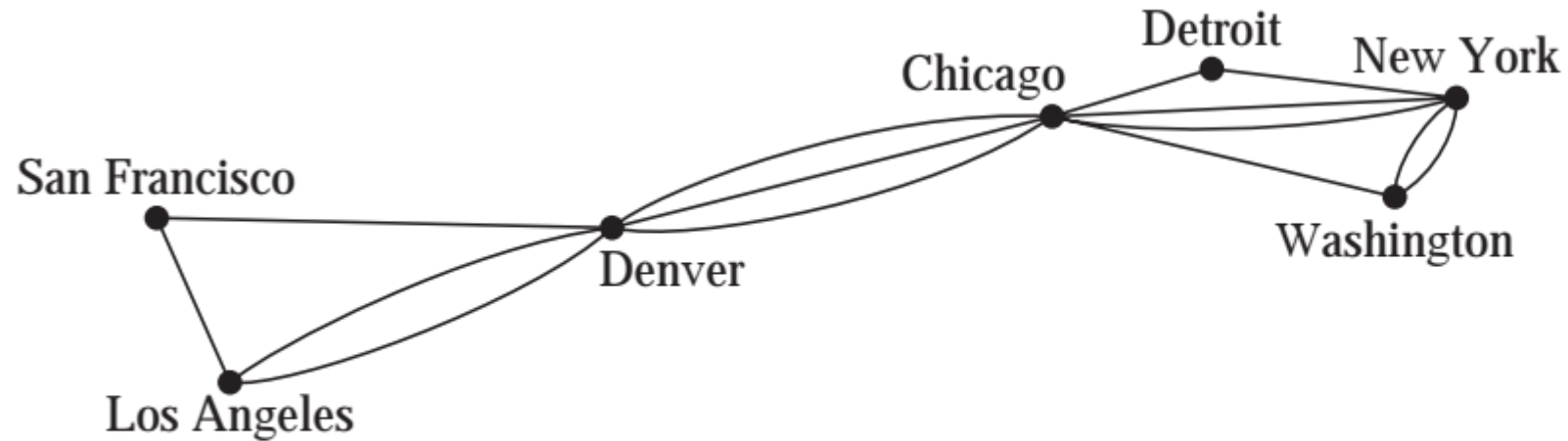


FIGURE 2 A Computer Network with Multiple Links between Data Centers.

- ❑ A computer network may contain multiple links between data centers, as shown in Figure 2.
- ❑ To model such networks, we need graphs that have more than one edge connecting the same pair of vertices

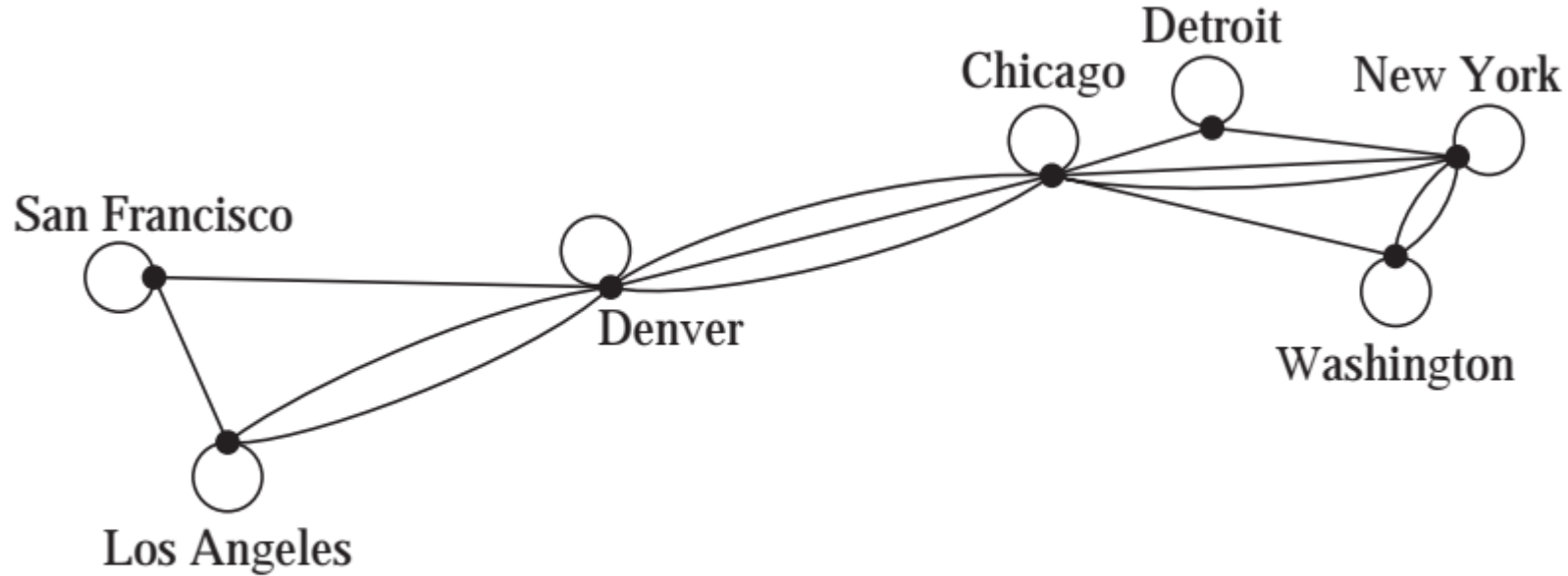


FIGURE 3 A Computer Network with Diagnostic Links.

- ❑ A communications link connects a data center with itself, perhaps a feedback loop for diagnostic purposes. Such a network is illustrated in Figure 3.

DEFINITION 2

A *directed graph* (or *digraph*) (V, E) consists of a nonempty set of vertices V and a set of *directed edges* (or *arcs*) E . Each directed edge is associated with an ordered pair of vertices. The directed edge associated with the ordered pair (u, v) is said to *start* at u and *end* at v .

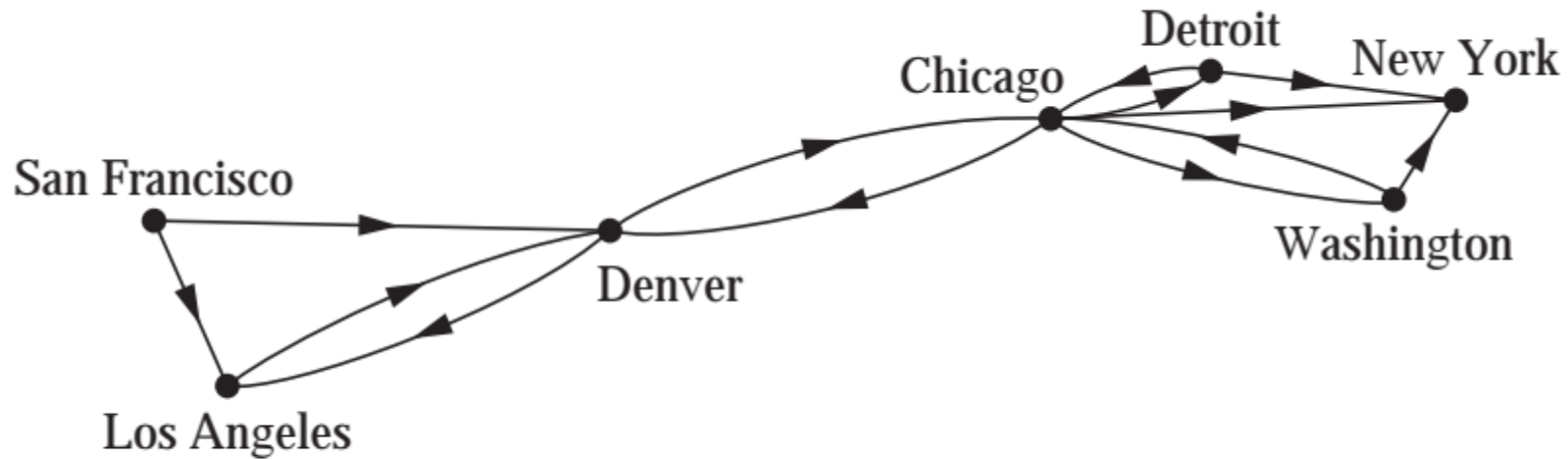


FIGURE 4 A Communications Network with One-Way Communications Links.

- This may be the case if there is a large amount of traffic sent to some data centers, with little or no traffic going in the opposite direction. Such a network is shown in Figure 4

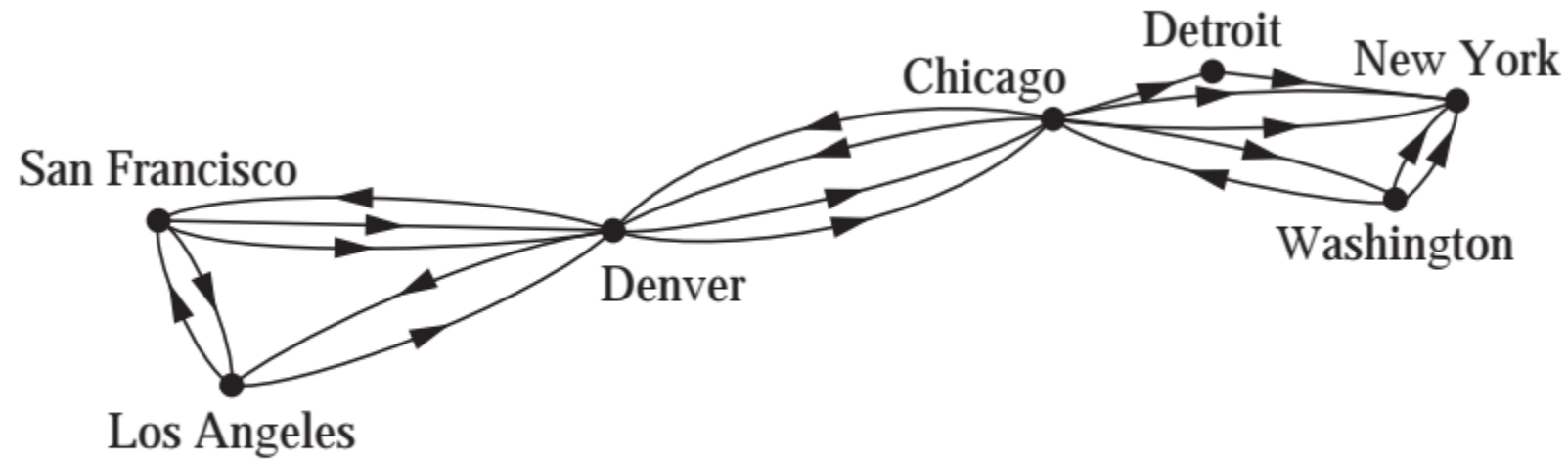


FIGURE 5 A Computer Network with Multiple One-Way Links.



- ❑ A graph in which each edge connects two different vertices and where no two edges connect the same pair of vertices is called a simple graph.
- ❑ Graphs that may have multiple edges connecting the same vertices are called multigraphs.
- ❑ Such edges are called loops, and sometimes we may even have more than one loop at a vertex
- ❑ Graphs that may include loops, and possibly multiple edges connecting the same pair of vertices or a vertex to itself, are sometimes called pseudographs.

TABLE 1 Graph Terminology.

<i>Type</i>	<i>Edges</i>	<i>Multiple Edges Allowed?</i>	<i>Loops Allowed?</i>
Simple graph	Undirected	No	No
Multigraph	Undirected	Yes	No
Pseudograph	Undirected	Yes	Yes
Simple directed graph	Directed	No	No
Directed multigraph	Directed	Yes	Yes
Mixed graph	Directed and undirected	Yes	Yes

Graph Model

❑ Social Networks

- Acquaintanceship and Friendship Graphs
- Influence Graphs
- Collaboration Graphs
- Call Graph

❑ INFORMATION NETWORKS

- The Web Graph
- Citation Graphs

❑ SOFTWARE DESIGN APPLICATIONS

- Module Dependency Graphs
- Precedence Graphs and Concurrent Processing

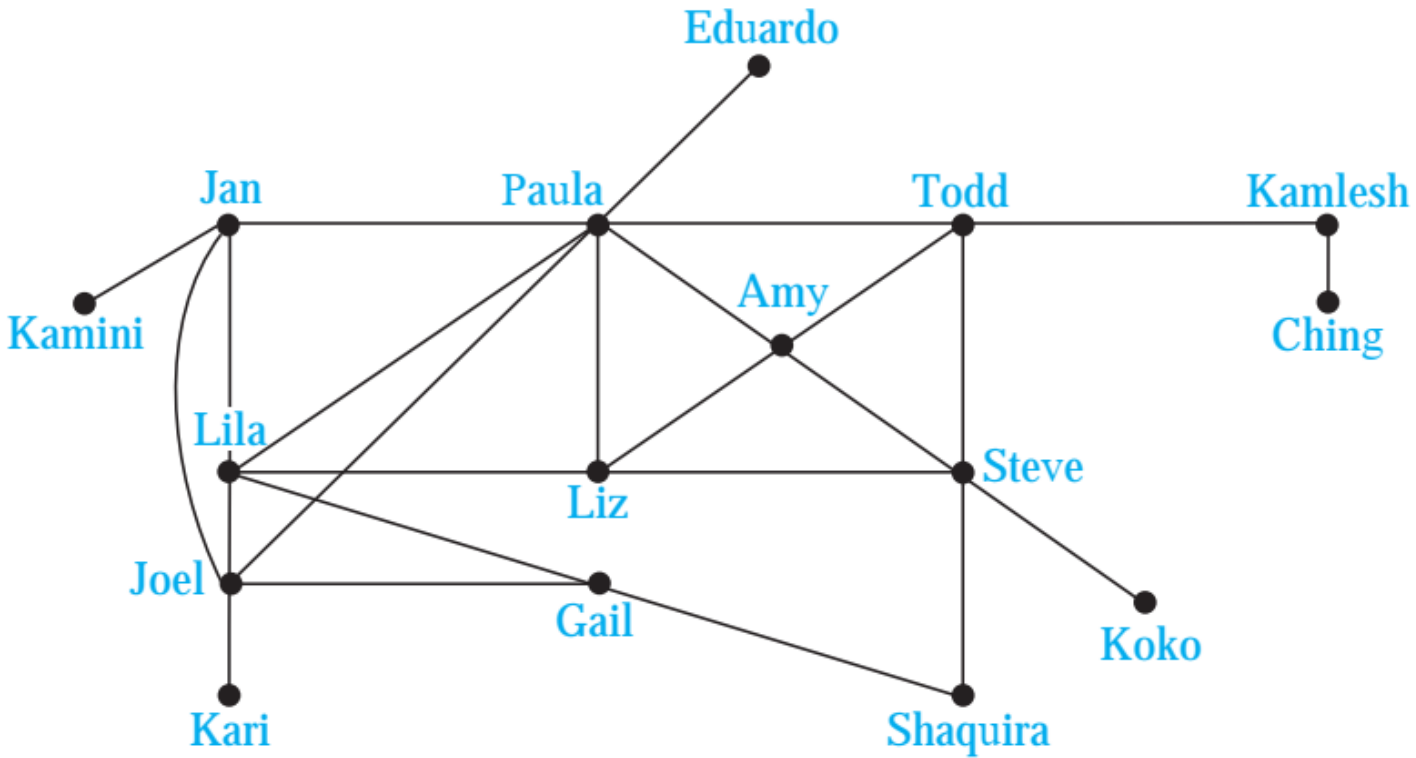


FIGURE 6 An Acquaintanceship Graph.

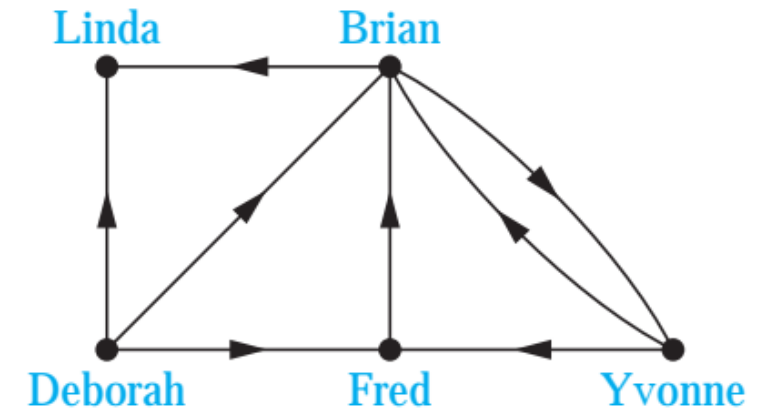


FIGURE 7 An Influence Graph.

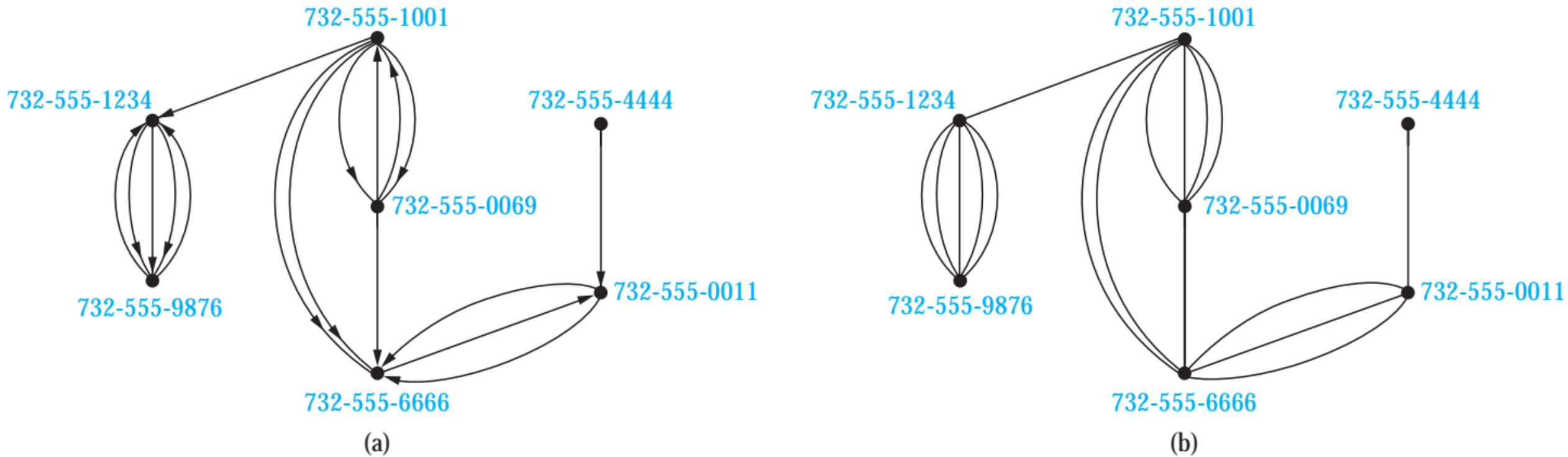


FIGURE 8 A Call Graph.

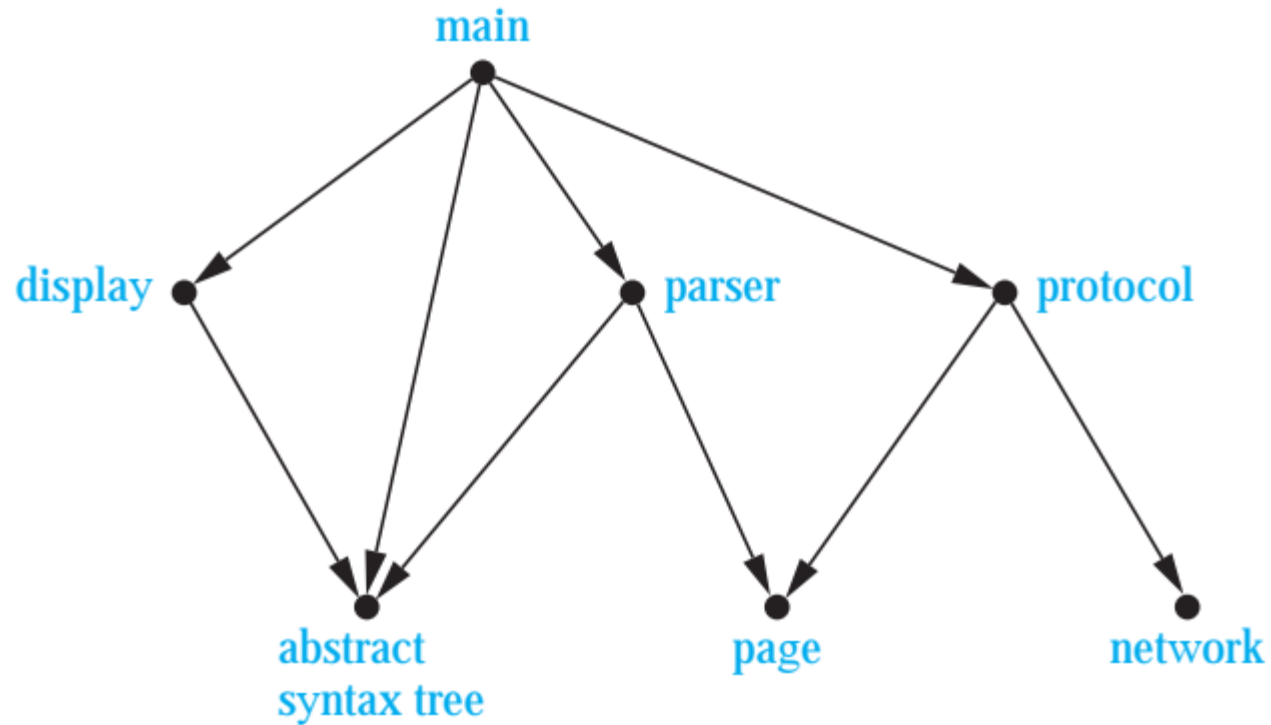


FIGURE 9 A Module Dependency Graph.

S_1 $a := 0$
 S_2 $b := 1$
 S_3 $c := a + 1$
 S_4 $d := b + a$
 S_5 $e := d + 1$
 S_6 $e := c + d$

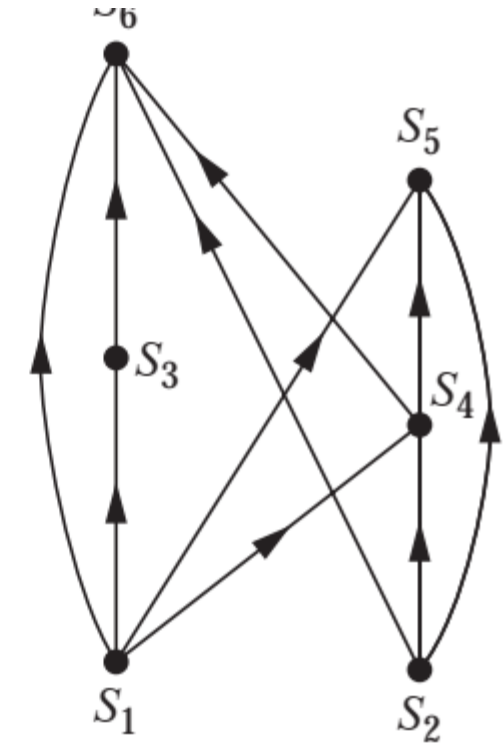


FIGURE 10 A Precedence Graph.

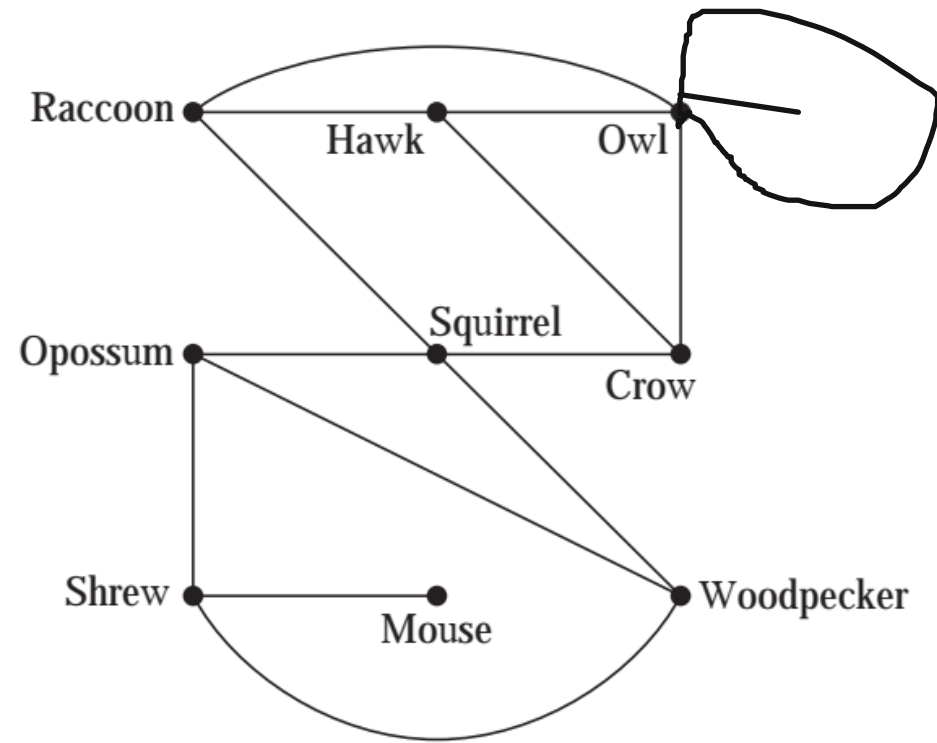


FIGURE 11 A Niche Overlap Graph.

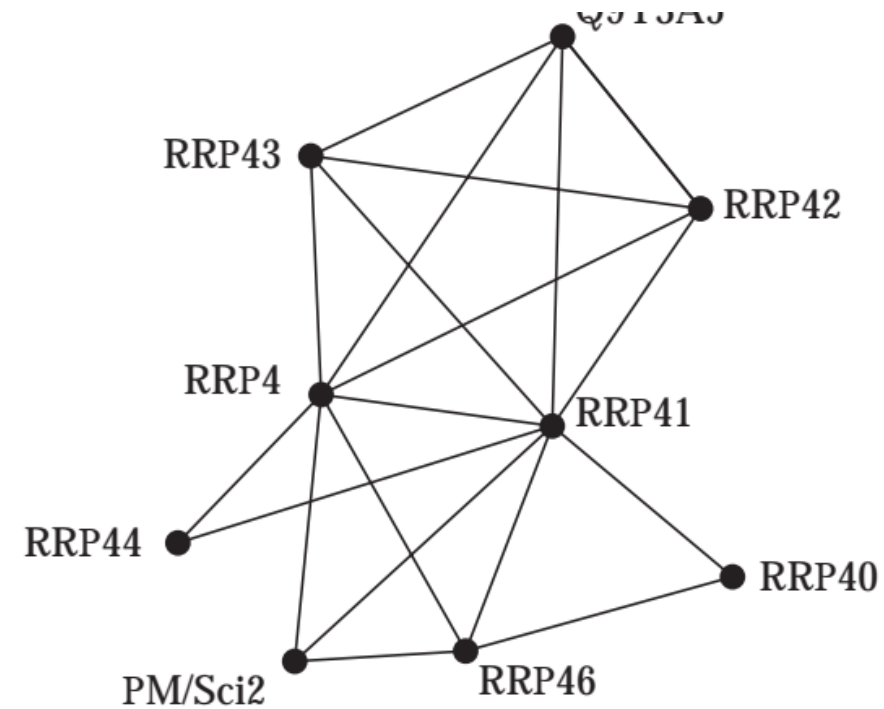


FIGURE 12 A Module of a Protein Interaction Graph.

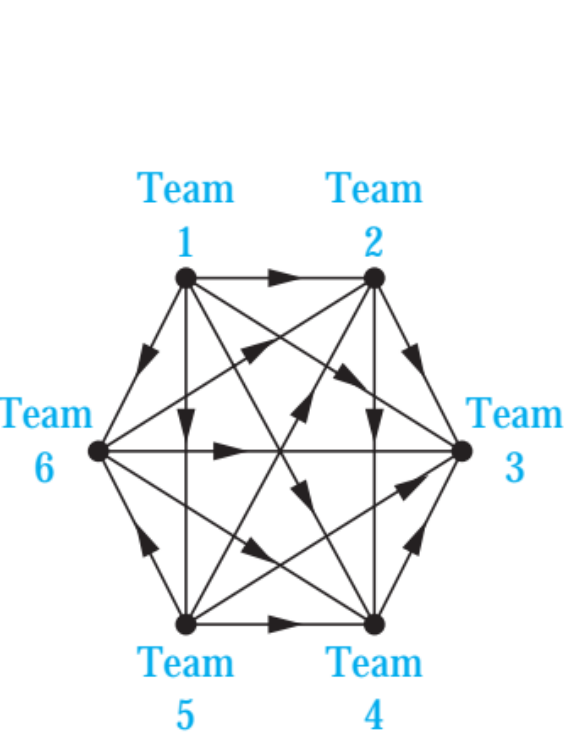


FIGURE 13 A Graph Model of a Round-Robin Tournament.

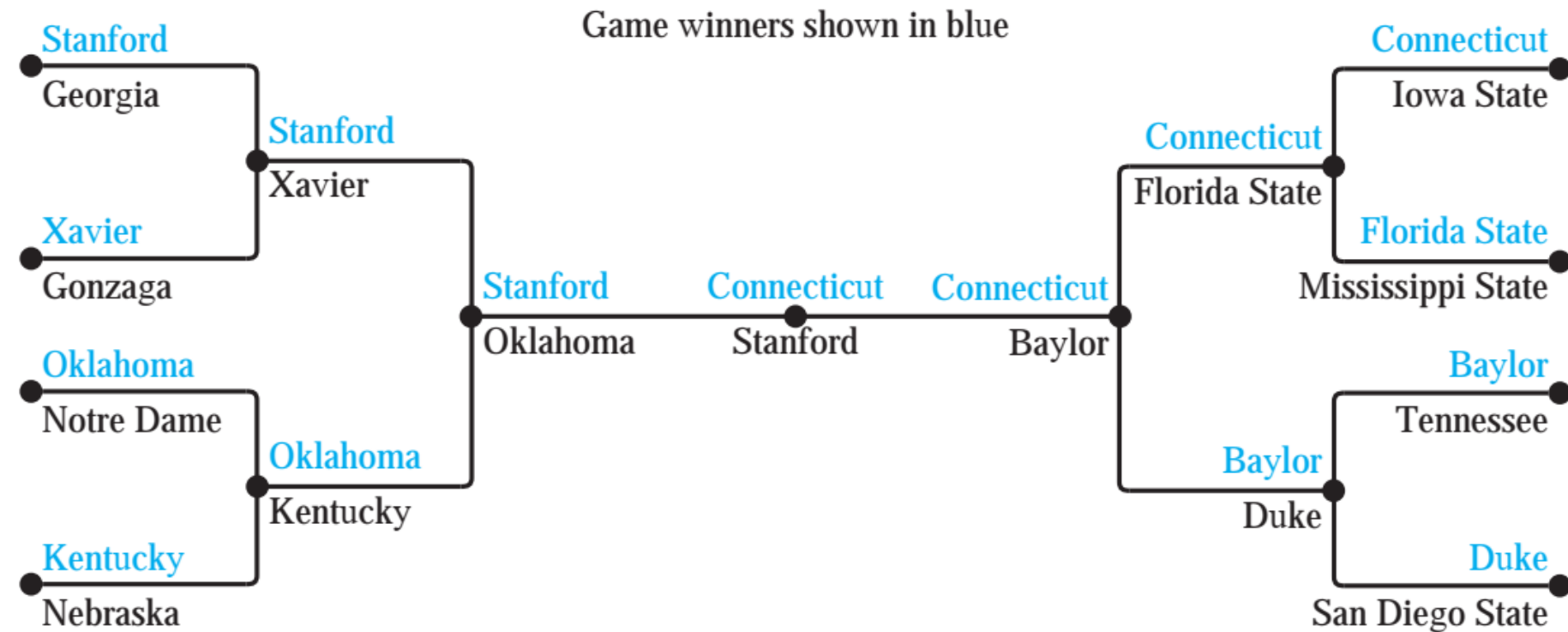


FIGURE 14 A Single-Elimination Tournament.

Exercises

1. Draw graph models, stating the type of graph (from Table 1) used, to represent airline routes where every day there are four flights from Boston to Newark, two flights from Newark to Boston, three flights from Newark to Miami, two flights from Miami to Newark, one flight from Newark to Detroit, two flights from Detroit to Newark, three flights from Newark to Washington, two flights from Washington to Newark, and one flight from Washington to Miami, with
 - a) an edge between vertices representing cities that have a flight between them (in either direction).
 - b) an edge between vertices representing cities for each flight that operates between them (in either direction).
 - c) an edge between vertices representing cities for each flight that operates between them (in either direction), plus a loop for a special sightseeing trip that takes off and lands in Miami.
 - d) an edge from a vertex representing a city where a flight starts to the vertex representing the city where it ends.
 - e) an edge for each flight from a vertex representing a city where the flight begins to the vertex representing the city where the flight ends.
2. What kind of graph (from Table 1) can be used to model a highway system between major cities where
 - a) there is an edge between the vertices representing cities if there is an interstate highway between them?
 - b) there is an edge between the vertices representing cities for each interstate highway between them?
 - c) there is an edge between the vertices representing cities for each interstate highway between them, and there is a loop at the vertex representing a city if there is an interstate highway that circles this city?

10.2 Graph Terminology and Special Types of Graphs

DEFINITION 1

Two vertices u and v in an undirected graph G are called *adjacent* (or *neighbors*) in G if u and v are endpoints of an edge e of G . Such an edge e is called *incident with* the vertices u and v and e is said to *connect* u and v .

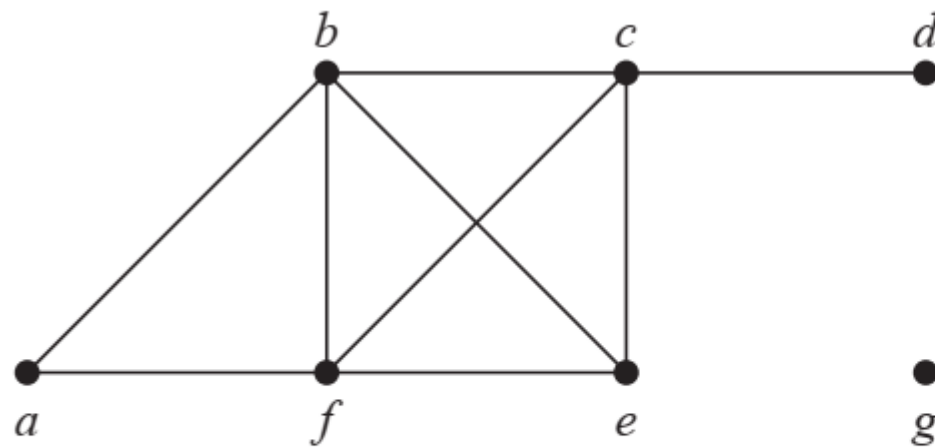
DEFINITION 2

The set of all neighbors of a vertex v of $G = (V, E)$, denoted by $N(v)$, is called the *neighborhood* of v . If A is a subset of V , we denote by $N(A)$ the set of all vertices in G that are adjacent to at least one vertex in A . So, $N(A) = \bigcup_{v \in A} N(v)$.

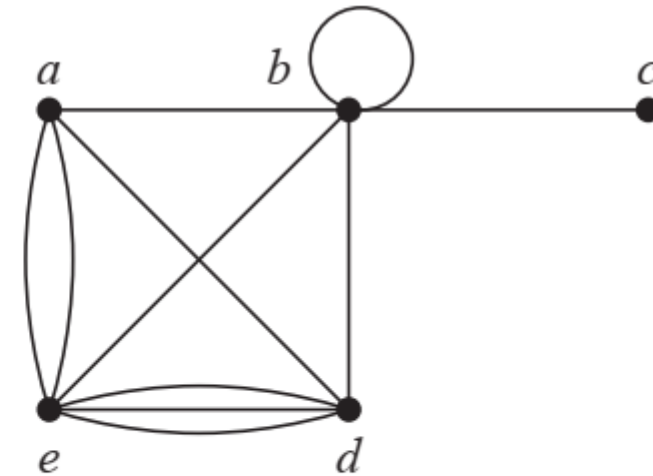
DEFINITION 3

The *degree of a vertex in an undirected graph* is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex. The degree of the vertex v is denoted by $\deg(v)$.

EXAMPLE 1 What are the degrees and what are the neighborhoods of the vertices in the graphs G and H displayed in Figure 1?



G



H

FIGURE 1 The Undirected Graphs G and H .

THEOREM 1

THE HANDSHAKING THEOREM Let $G = (V, E)$ be an undirected graph with m edges. Then

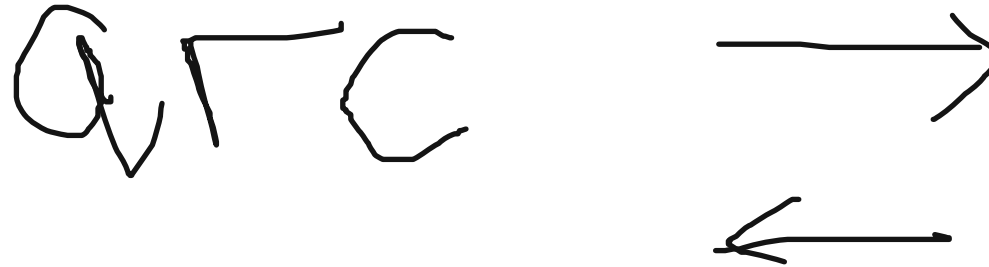
$$2m = \sum_{v \in V} \deg(v).$$

(Note that this applies even if multiple edges and loops are present.)

EXAMPLE 3 How many edges are there in a graph with 10 vertices each of degree six?

THEOREM 2

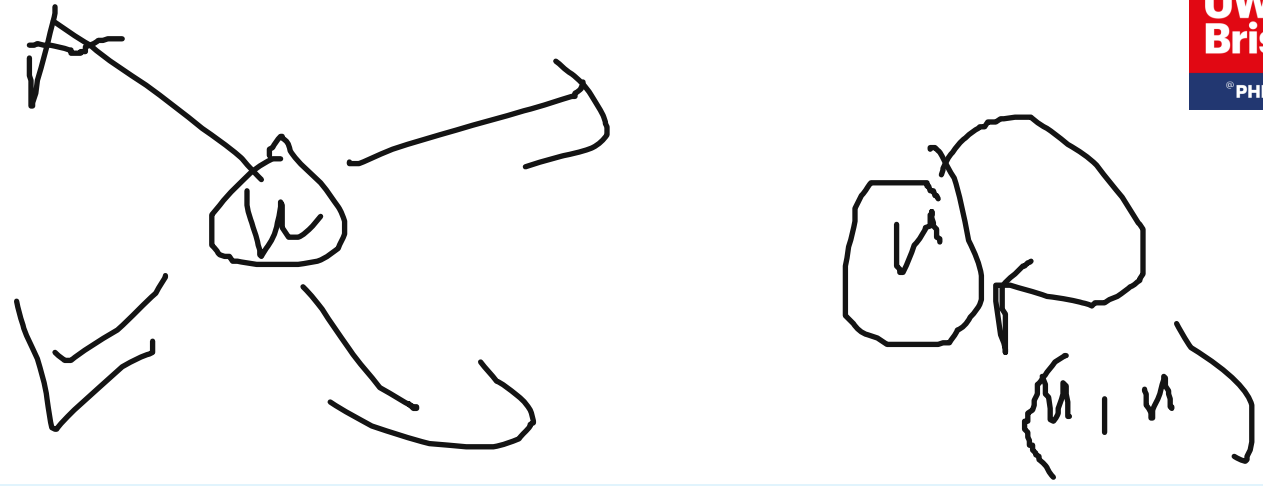
An undirected graph has an even number of vertices of odd degree.



DEFINITION 4

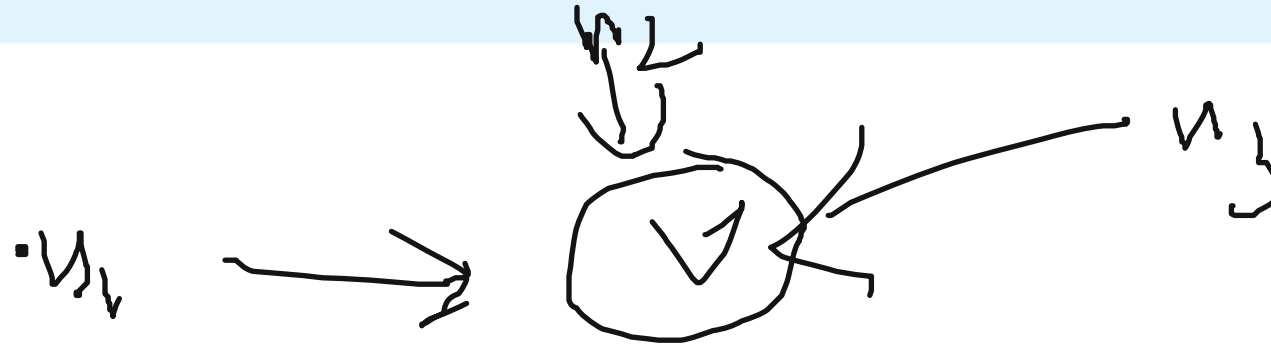
When (u, v) is an edge of the graph G with directed edges, u is said to be *adjacent to* v and v is said to be *adjacent from* u . The vertex u is called the *initial vertex* of (u, v) , and v is called the *terminal* or *end vertex* of (u, v) . The initial vertex and terminal vertex of a loop are the same.





DEFINITION 5

In a graph with directed edges the in-degree of a vertex v , denoted by $\deg^-(v)$, is the number of edges with v as their terminal vertex. The out-degree of v , denoted by $\deg^+(v)$, is the number of edges with v as their initial vertex. (Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of this vertex.)



EXAMPLE 4

Find the in-degree and out-degree of each vertex in the graph G with directed edges shown in Figure 2.

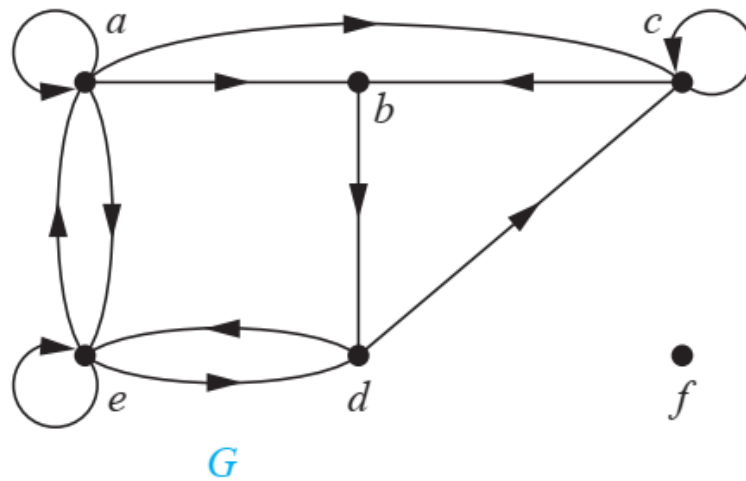


FIGURE 2 The Directed Graph G .

THEOREM 3

Let $G = (V, E)$ be a graph with directed edges. Then

$$\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|.$$

EXAMPLE 5 Complete Graphs A **complete graph on n vertices**, denoted by K_n , is a simple graph that contains exactly one edge between each pair of distinct vertices. The graphs K_n , for $n = 1, 2, 3, 4, 5, 6$, are displayed in Figure 3. A simple graph for which there is at least one pair of distinct vertex not connected by an edge is called **noncomplete**.

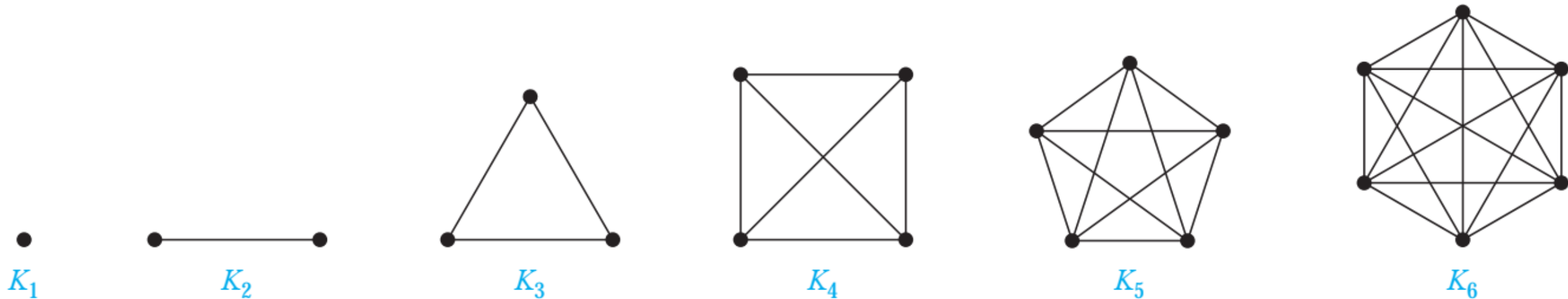


FIGURE 3 The Graphs K_n for $1 \leq n \leq 6$.

EXAMPLE 6 **Cycles** A cycle C_n , $n \geq 3$, consists of n vertices $v_1, v_2, \dots, v_n$ and edges $\{v_1, v_2\}$, $\{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}$, and $\{v_n, v_1\}$. The cycles C_3 , C_4 , C_5 , and C_6 are displayed in Figure 4.

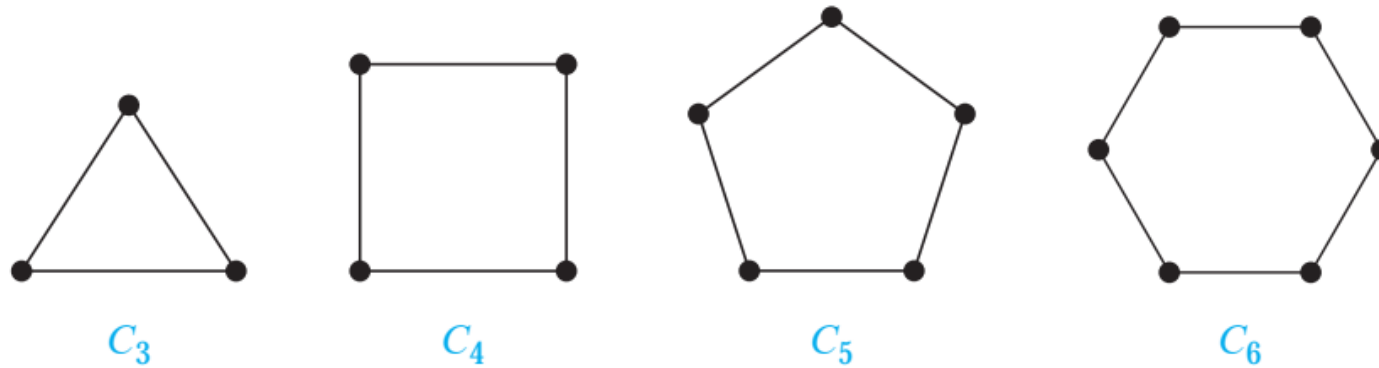


FIGURE 4 The Cycles C_3 , C_4 , C_5 , and C_6 .

EXAMPLE 7

Wheels We obtain a **wheel** W_n when we add an additional vertex to a cycle C_n , for $n \geq 3$, and connect this new vertex to each of the n vertices in C_n , by new edges. The wheels W_3 , W_4 , W_5 , and W_6 are displayed in Figure 5.

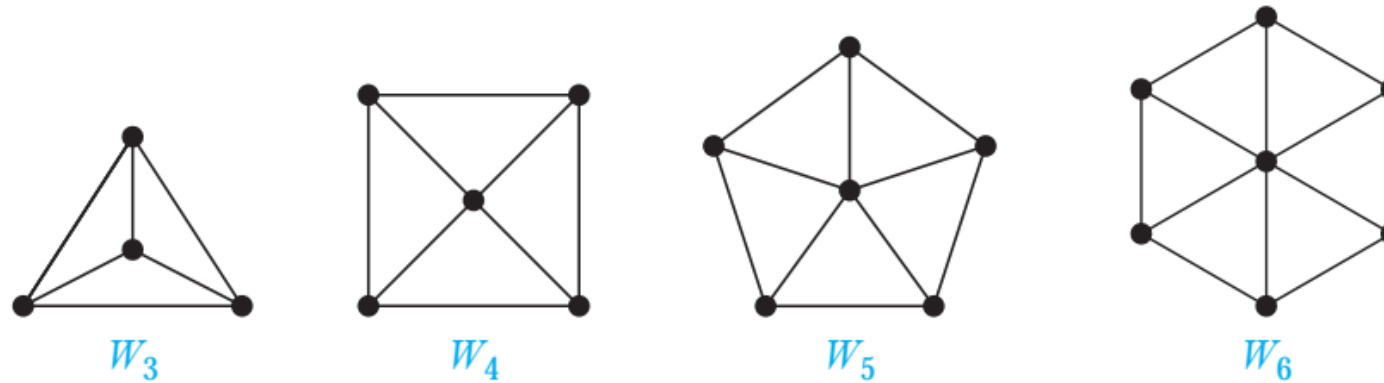


FIGURE 5 The Wheels W_3 , W_4 , W_5 , and W_6 .

10.3 Representing Graphs and Graph Isomorphism

- ❑ List all the edges of this graph.
- ❑ Use adjacency lists, which specify the vertices that are adjacent to each vertex of the graph
- ❑ Matrices
 - One is based on the adjacency of vertices,
 - The other is based on incidence of vertices and edges.

EXAMPLE 1 Use adjacency lists to describe the simple graph given in Figure 1.

Solution: Table 1 lists those vertices adjacent to each of the vertices of the graph.

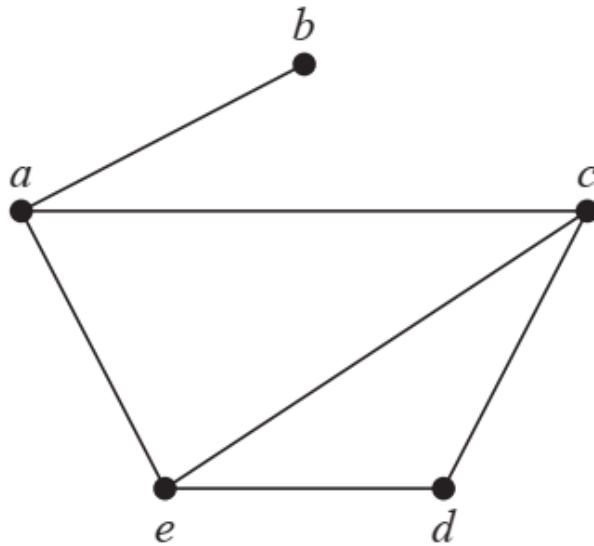


FIGURE 1 A Simple Graph.

TABLE 1 An Adjacency List for a Simple Graph.

<i>Vertex</i>	<i>Adjacent Vertices</i>
<i>a</i>	<i>b, c, e</i>
<i>b</i>	<i>a</i>
<i>c</i>	<i>a, d, e</i>
<i>d</i>	<i>c, e</i>
<i>e</i>	<i>a, c, d</i>

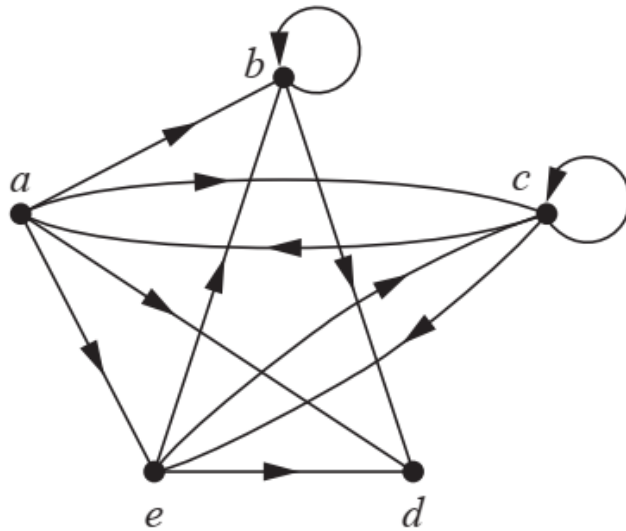


FIGURE 2 A Directed Graph.

TABLE 2 An Adjacency List for a Directed Graph.

<i>Initial Vertex</i>	<i>Terminal Vertices</i>
<i>a</i>	<i>b, c, d, e</i>
<i>b</i>	<i>b, d</i>
<i>c</i>	<i>a, c, e</i>
<i>d</i>	
<i>e</i>	<i>b, c, d</i>

EXAMPLE 2

Represent the directed graph shown in Figure 2 by listing all the vertices that are the terminal vertices of edges starting at each vertex of the graph.

Adjacency Matrices

Suppose that $G = (V, E)$ is a simple graph where $|V| = n$. Suppose that the vertices of G are listed arbitrarily as $v_1, v_2, \dots, v_n$. The **adjacency matrix** $\mathbf{A}$ (or $\mathbf{A}_G$) of G , with respect to this listing of the vertices, is the $n \times n$ zero-one matrix with 1 as its (i, j) th entry when v_i and v_j are adjacent, and 0 as its (i, j) th entry when they are not adjacent. In other words, if its adjacency matrix is $\mathbf{A} = [a_{ij}]$, then

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G, \\ 0 & \text{otherwise.} \end{cases}$$

EXAMPLE 3 Use an adjacency matrix to represent the graph shown in Figure 3.

Solution: We order the vertices as a, b, c, d . The matrix representing this graph is

$$\begin{array}{c} a \\ b \\ c \\ d \end{array} \begin{array}{c} a \quad b \quad c \quad d \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

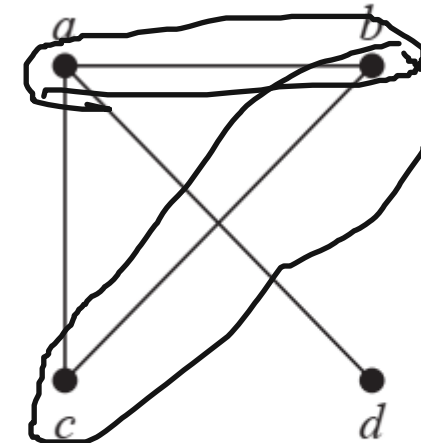
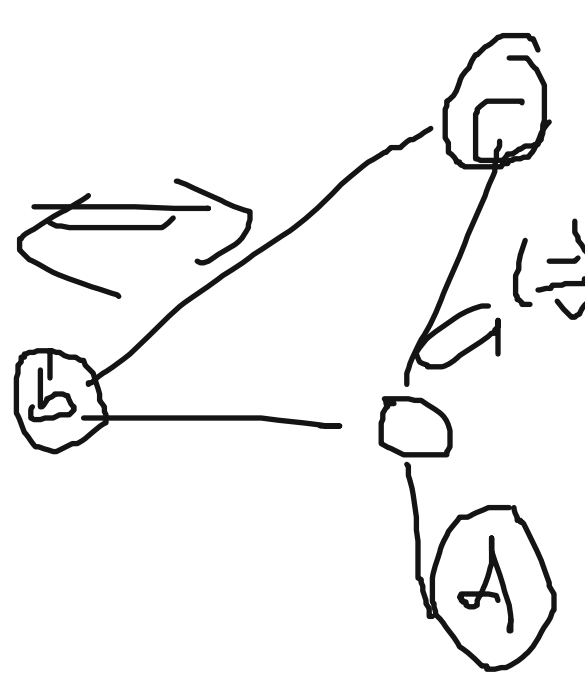


FIGURE 3
Simple Graph.

EXAMPLE 5 Use an adjacency matrix to represent the pseudograph shown in Figure 5.

Solution: The adjacency matrix using the ordering of vertices a, b, c, d is

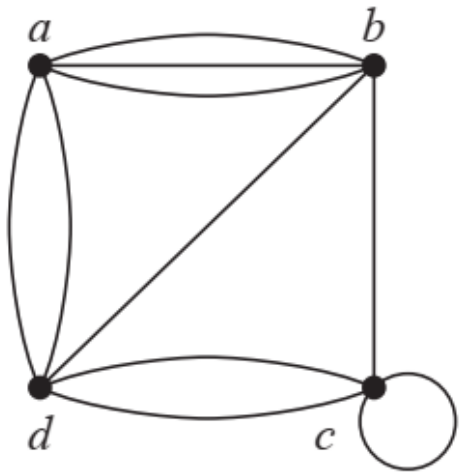


FIGURE 5
A Pseudograph.

$$\begin{bmatrix} 0 & 3 & 0 & 2 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix} \cdot \text{h} \times \text{W}$$

We used zero-one matrices in Chapter 9 to represent directed graphs. The matrix for a directed graph $G = (V, E)$ has a 1 in its (i, j) th position if there is an edge from v_i to v_j , where $v_1, v_2, \dots, v_n$ is an arbitrary listing of the vertices of the directed graph. In other words, if $\mathbf{A} = [a_{ij}]$ is the adjacency matrix for the directed graph with respect to this listing of the vertices, then

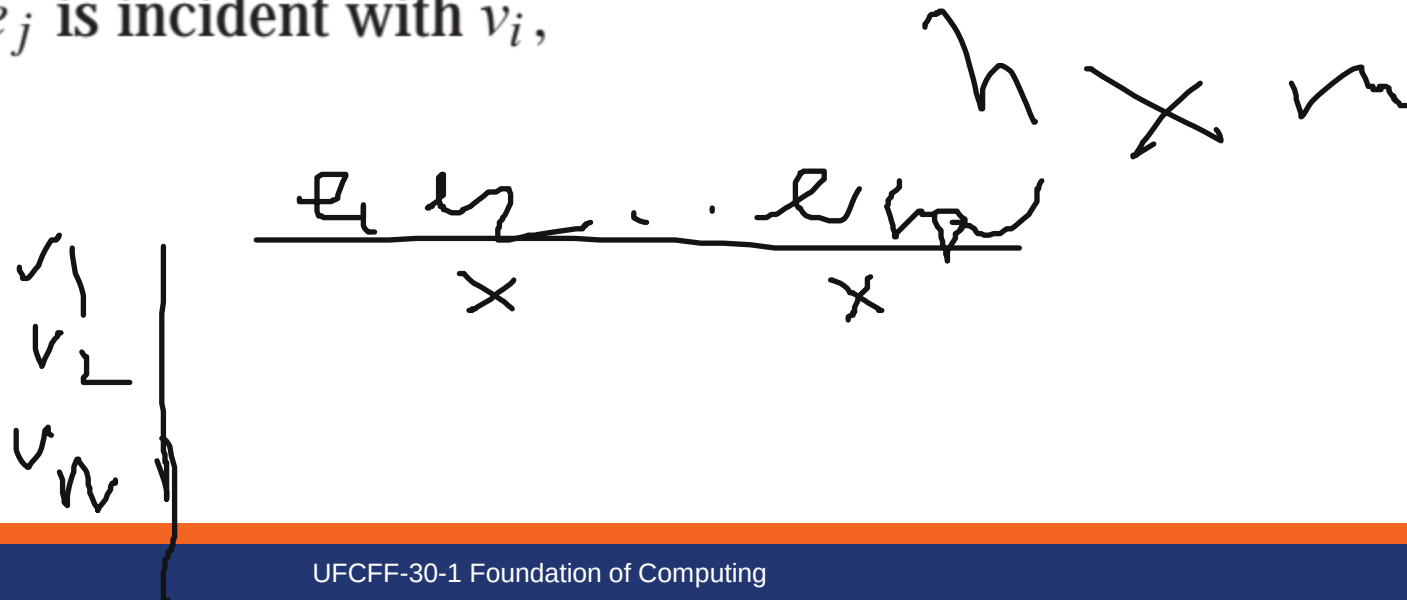
$$a_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \text{ is an edge of } G, \\ 0 & \text{otherwise.} \end{cases}$$

The adjacency matrix for a directed graph does not have to be symmetric, because there may not be an edge from v_j to v_i when there is an edge from v_i to v_j .

Incidence Matrices

Another common way to represent graphs is to use **incidence matrices**. Let $G = (V, E)$ be an undirected graph. Suppose that $v_1, v_2, \dots, v_n$ are the vertices and $e_1, e_2, \dots, e_m$ are the edges of G . Then the incidence matrix with respect to this ordering of V and E is the $n \times m$ matrix $\mathbf{M} = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i, \\ 0 & \text{otherwise.} \end{cases}$$



EXAMPLE 6 Represent the graph shown in Figure 6 with an incidence matrix.

Solution: The incidence matrix is

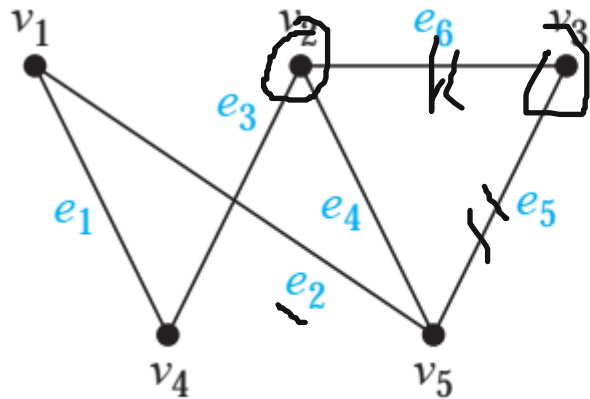
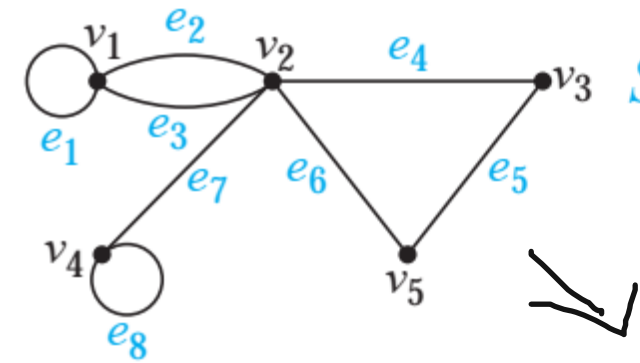


FIGURE 6 An
Undirected
Graph.

$$\approx \begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

sparse
table

EXAMPLE 7 Represent the pseudograph shown in Figure 7 using an incidence matrix.



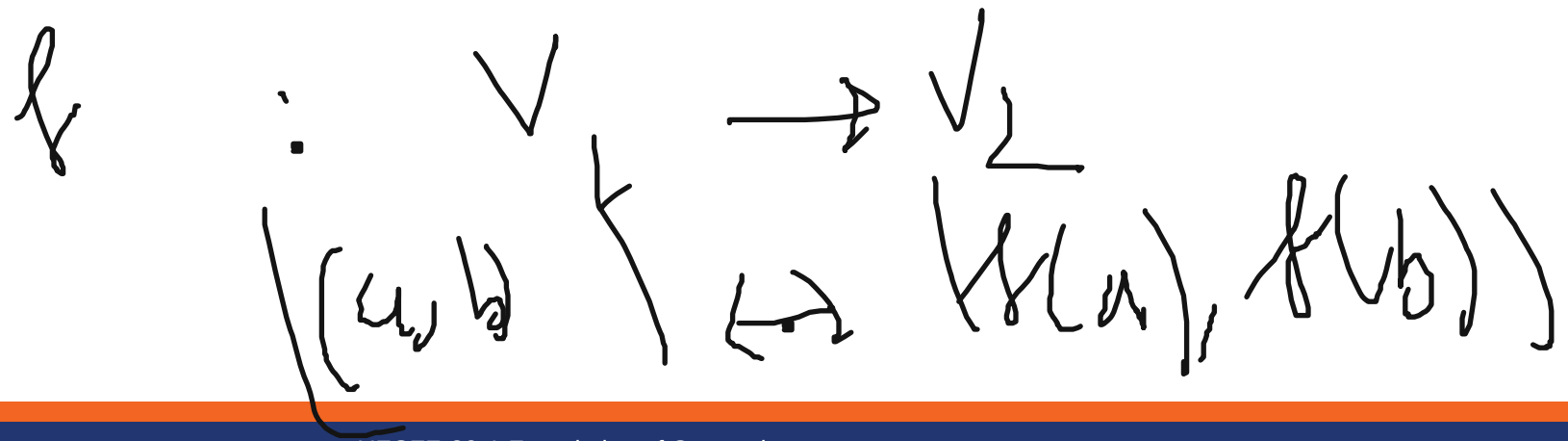
Solution: The incidence matrix for this graph is

$$\begin{array}{c}
 v_1 \\
 v_2 \\
 v_3 \\
 v_4 \\
 v_5
 \end{array}
 \begin{array}{c}
 e_1 \quad e_2 \quad e_3 \quad e_4 \quad e_5 \quad e_6 \quad e_7 \quad e_8 \\
 \left[\begin{array}{cccccccc}
 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\
 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\
 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\
 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0
 \end{array} \right]
 \end{array}
 .$$

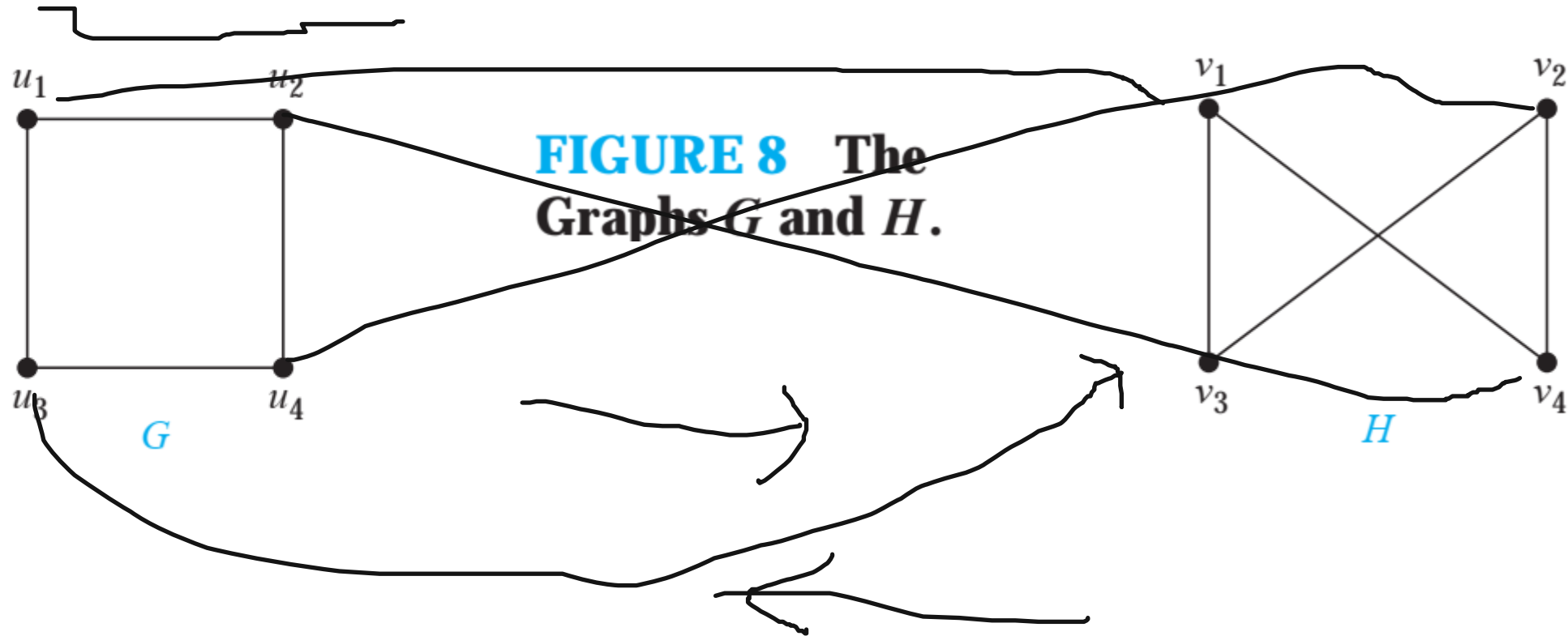
FIGURE 7
A Pseudograph.

DEFINITION 1

The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there exists a one-to-one and onto function f from V_1 to V_2 with the property that a and b are adjacent in G_1 if and only if $f(a)$ and $f(b)$ are adjacent in G_2 , for all a and b in V_1 . Such a function f is called an isomorphism.* Two simple graphs that are not isomorphic are called *nonisomorphic*.



EXAMPLE 8 Show that the graphs $G = (V, E)$ and $H = (W, F)$, displayed in Figure 8, are isomorphic.



EXAMPLE 9 Show that the graphs displayed in Figure 9 are not isomorphic.

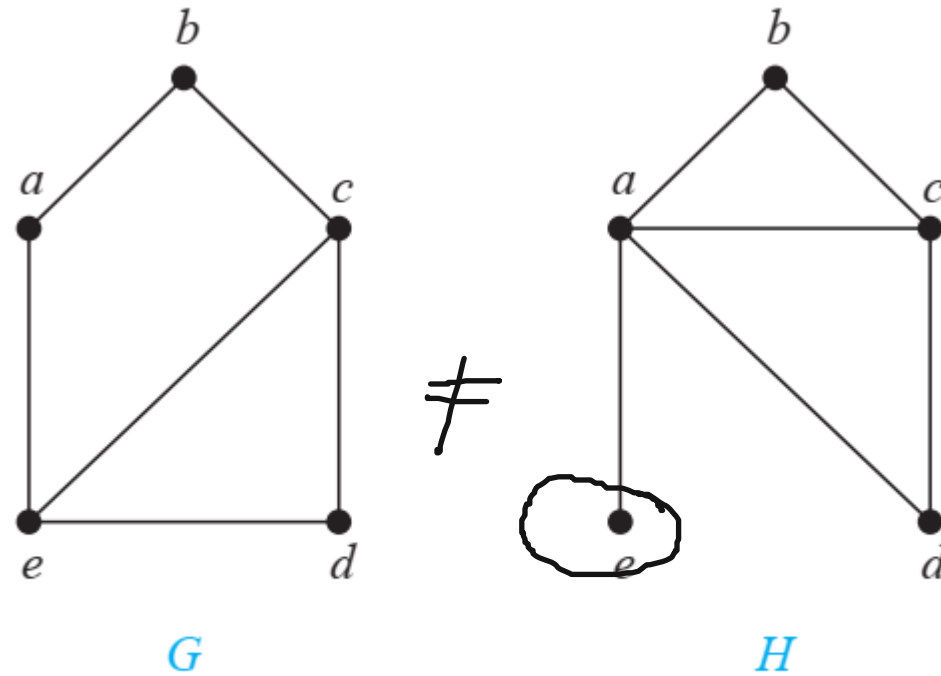


FIGURE 9 The Graphs G and H .

10.4 Connectivity

- Many problems can be modeled with paths formed by traveling along the edges of graphs

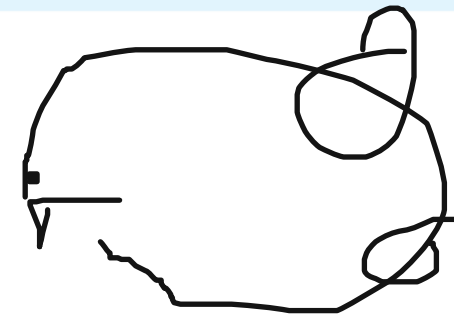
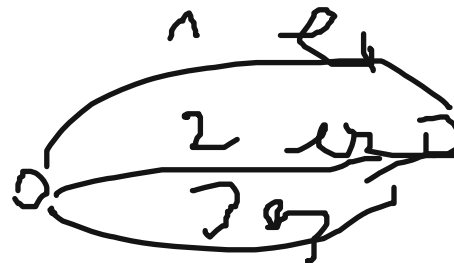


Paths

Informally, a **path** is a sequence of edges that begins at a vertex of a graph and travels from vertex to vertex along edges of the graph. As the path travels along its edges, it visits the vertices along this path, that is, the endpoints of these edges.

DEFINITION 1

Let n be a nonnegative integer and G an undirected graph. A *path* of length n from u to v in G is a sequence of n edges $e_1, \dots, e_n$ of G for which there exists a sequence $x_0 = u, x_1, \dots, x_{n-1}, x_n = v$ of vertices such that e_i has, for $i = 1, \dots, n$, the endpoints x_{i-1} and x_i . When the graph is simple, we denote this path by its vertex sequence $x_0, x_1, \dots, x_n$ (because listing these vertices uniquely determines the path). The path is a *circuit* if it begins and ends at the same vertex, that is, if $u = v$, and has length greater than zero. The path or circuit is said to *pass through* the vertices $x_1, x_2, \dots, x_{n-1}$ or *traverse* the edges $e_1, e_2, \dots, e_n$. A path or circuit is *simple* if it does not contain the same edge more than once.



EXAMPLE 1

In the simple graph shown in Figure 1, a, d, c, f, e is a simple path of length 4, because $\{a, d\}$, $\{d, c\}$, $\{c, f\}$, and $\{f, e\}$ are all edges. However, ~~d, e, c, a is not a path~~, because $\{e, c\}$ is not an edge. Note that b, c, f, e, b is a circuit of length 4 because $\{b, c\}$, $\{c, f\}$, $\{f, e\}$, and $\{e, b\}$ are edges, and this path begins and ends at b . The path a, b, e, d, a, b , which is of length 5, is not simple because it contains the edge $\{a, b\}$ twice.

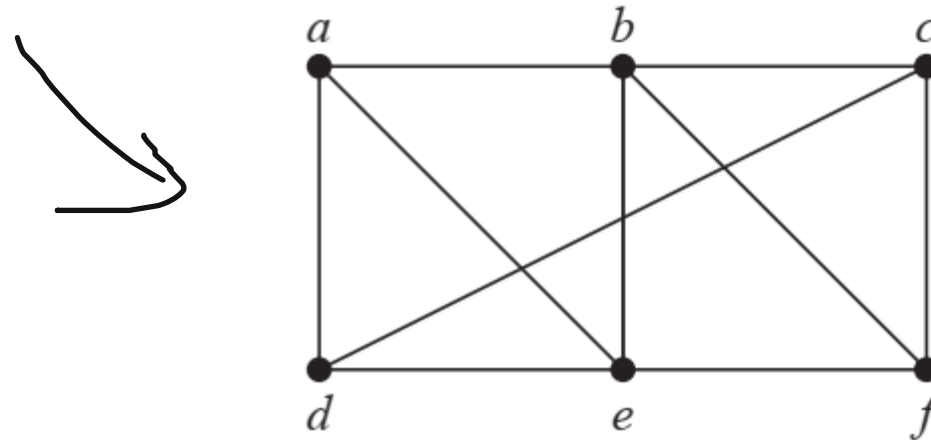


FIGURE 1 A Simple Graph.

DEFINITION 2

Let n be a nonnegative integer and G a directed graph. A *path* of length n from u to v in G is a sequence of edges $e_1, e_2, \dots, e_n$ of G such that e_1 is associated with (x_0, x_1) , e_2 is associated with (x_1, x_2) , and so on, with e_n associated with (x_{n-1}, x_n) , where $x_0 = u$ and $x_n = v$. When there are no ~~multiple edges in the directed graph~~, this path is denoted by its vertex sequence $x_0, x_1, x_2, \dots, x_n$. A path of length greater than zero that begins and ends at the same vertex is called a *circuit* or *cycle*. A path or circuit is called *simple* if it does not contain the same edge more than once.

Connectedness in Undirected Graphs

- When does a computer network have the property that every pair of computers can share information, if messages can be sent through one or more intermediate computers

DEFINITION 3

An undirected graph is called *connected* if there is a path between every pair of distinct vertices of the graph. An undirected graph that is not *connected* is called *disconnected*. We say that we *disconnect* a graph when we remove vertices or edges, or both, to produce a disconnected subgraph.

EXAMPLE 4 The graph G_1 in Figure 2 is connected, because for every pair of distinct vertices there is a path between them (the reader should verify this). However, the graph G_2 in Figure 2 is not connected. For instance, there is no path in G_2 between vertices a and d .

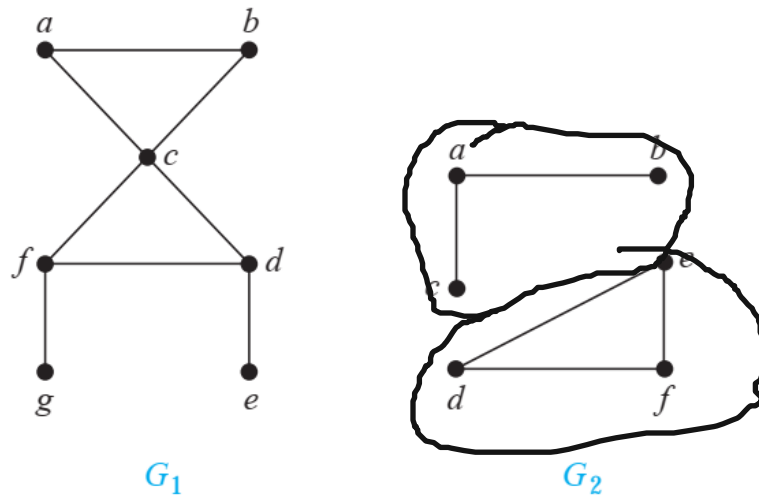


FIGURE 2 The Graphs G_1 and G_2 .

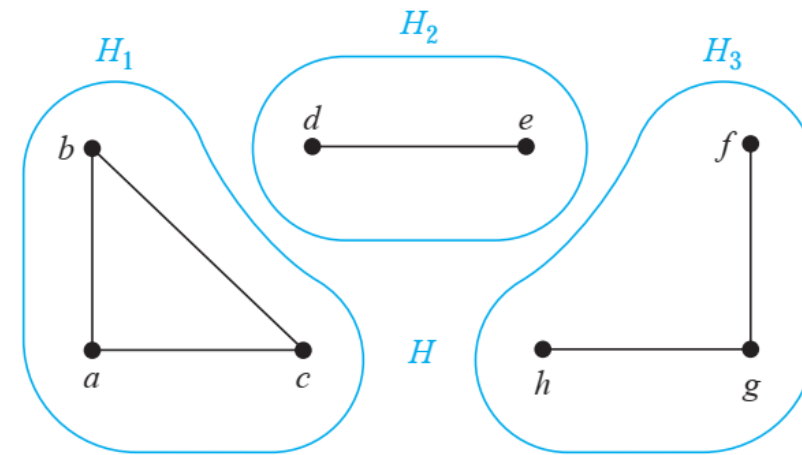


FIGURE 3 The Graph H and Its Connected Components H_1, H_2 , and H_3 .

Counting Paths Between Vertices

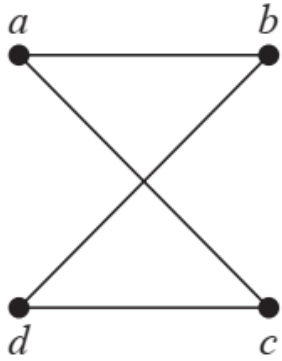
$$\frac{v_i \times v_j}{\text{---}}$$

THEOREM 2

Let G be a graph with adjacency matrix $\mathbf{A}$ with respect to the ordering $v_1, v_2, \dots, v_n$ of the vertices of the graph (with directed or undirected edges, with multiple edges and loops allowed). The number of different paths of length r from v_i to v_j , where r is a positive integer, equals the (i, j) th entry of $\mathbf{A}^r$.

$$\mathbf{A}^r = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

EXAMPLE 15 How many paths of length four are there from a to d in the simple graph G in Figure 8?



Solution: The adjacency matrix of G (ordering the vertices as a, b, c, d) is

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}.$$

FIGURE 8 The Graph G .

Hence, the number of paths of length four from a to d is the $(1, 4)$ th entry of $\mathbf{A}^4$. Because

$$\mathbf{A}^4 = \begin{bmatrix} 8 & 0 & 0 & 8 \\ 0 & 8 & 8 & 0 \\ 0 & 8 & 8 & 0 \\ 8 & 0 & 0 & 8 \end{bmatrix},$$

Conclusions

- ❑ Order and size
- ❑ Degree and degree sequence
- ❑ Walk, path and cycle
- ❑ Isomorphic graphs
- ❑ Graph representation